Krishna Agarwal

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CPI: **9.57**/10 (IIT Bombay, '28) | Intermediate: **92.80%** (CBSE, '24) | Matriculation: **97.00%** (CBSE, '22)

Pursuing **Dual Minor** in **Computer Science** and **Artificial Intelligence & Data Science** at IIT Bombay

Scholastic Achievements

- Awarded **AA grade(10/10)** in **15** courses including **Calculus** among **1300+** students [2026]
- Secured **AIR 531**, acknowledged as **state topper** in West Bengal, **National Standard Exam in Chemistry** [2023]
- Secured a percentile of **99.91** in **JEE Mains** Examination out of **1.4 million+** participants [2024]
- Secured an **AIR** of **1042** in **JEE Advance** Examination out of **0.18 million+** participants [2024]
- Scored **319/390** (among the top performers being offered a seat at **BITS Pilani**) in **BITSAT** [2024]

Professional Experience

AI Research Intern | **Godrej Consumer Products Limited** [May '26 - Jul '26]

- Architected **3-layer**, **30+** step Agentic-AI foundation from scratch for Godrej, built reusable for future AI tools
- Built **15-step** LLM pipeline scoring signals across **6** FMCG rivals: HUL, P&G, ITC, Marico, Dabur, Reckitt
- Designed **6** modular source adapters on a swappable local to Azure layer spanning queue, blob, and LLM providers
- Shipped a **40+** table data model with self-governing feedback loop and a **7-router** reusable REST API

Software Development Intern | **DevopsAgents** [Jun '25 - Jul '25]

*Earned a **Letter of Recommendation** from the founder for delivering **3 times** higher savings than prior attempts*

- Reduced token usage costs by **70%** through implementing **prompt caching** for **LLM** based DevOps agents
- Manipulated **LangChain** classes to integrate with **OpenRouter**, enabling prompt caching **data retrieval**

Competitions

Intelligent Cart Super Add-On (CSAO) | **Zomathon Data Hackathon** | **Eternal** [Mar '26 - Apr '26]

*Secured **2nd/6,000+** place **nationally**, developing a real-time, context-aware cart add-on recommender system*

- Built **4**-level synthesis pipeline with **10K+** orders via **3** realism injections mirroring stochastic real-world scenarios
- Engineered a hybrid neural architecture fusing **Set Transformer** and **GRU4Rec** via cross-attention with **LightGBM**
- Designed **3**-tier cold-start strategy leveraging SLM embeddings, LLM knowledge graphs and Bayesian priors
- Achieved ∼**65ms** P99 latency (3× below SLA) via asynchronous ONNX execution and a **Redis**-backed pre-fetch layer

Multi-Agent Investment Teaser Generator | **AI-ML Hackathon** | **Kelp Global** [Feb '26 - Mar '26]

***1st/400+** at IITB: Built autonomous multi-agent system generating consulting-grade M&A pitch decks*

- Designed a **6-agent** workflow to autonomously scrape public web data from Google and fuse private financials
- Engineered **JSON schemas** prompting Gemini to autonomously orchestrate sector-adaptive slide logic and content
- Created highly interactive **Next.js** UI featuring a **Slate.js** rich-text editor and dynamic charts for seamless visualization
- Generated fully editable, brand-compliant **.pptx** files in under **5 min** by integrating **Zustand** and **PptxGenJS**

Key Projects

Deep Reinforcement Learning for AlgoTrading | **Finsearch** | **Finance Club** [Jun '25 - Aug '25]

- **15.3%** out-of-sample return achieved over **6** weeks by building and deploying a **Deep RL (PPO)** trading agent
- **150%** annualized returns achieved, outperforming ARIMA, LSTM, and Buy-and-Hold benchmark strategies
- **Sharpe 2.13**, **Sortino 2.76**, and **Max Drawdown -2.89%** validated through t-tests and bootstrap confidence intervals
- Engineered a **risk-aware** trading framework using RSI, MACD, momentum, volatility, and reward optimization

Exact MILP Solver using Distributed Optimization | **Prof. Avinash Bhardwaj** [Jan '26 - Apr '26]

- Engineered an exact solver **27-106x** faster than Gurobi (N=100-10,000) via a min-plus convolution reduction
- Proved the convex regime collapses to an O(N log N) merge, verified vs Gurobi (**16/16**) and exhaustive truth (**12,000+**)
- Mapped the weight-depth phase boundary over **5M+** instances, locating where coupling forces exponential runtime

Gen-AI for Healthcare Reports | **Seasons of Code** | **WnCC IIT Bombay** [May '25 - Jul '25]

- Built an end-to-end **GenAI medical chatbot** with **LangChain** on a **1200+** page medical dataset
- Implemented processing of **symptom-based inputs** to generate **structured medical reports**
- Implemented **semantic search** and a **RAG** pipeline with **Hugging Face embeddings** and **Pinecone**
- Developed modular **Python backend** and **Flask UI** with proper version control using **GitHub**

Insti-Connect | **DevCom** | **IIT Bombay** [Jan '25 - Feb '25]

- Worked in a team of **6** developers to develop a LinkedIn-style **social media app** for IIT Bombay students
- Developed **18+ APIs** in a backend module made using **Django, Django REST framework, Simple-Jwt**

Stock Market Prediction | **Course Project** | **Probability and Statistics** [Jan '25 - Apr '25]

Guide: Prof. Narayan Rangaraj, Dept. of Industrial Engineering and Operations Research

- Achieved **96.93%** trend prediction accuracy by building a **Python-based** model using **moving averages**
- Developed a robust **Monte Carlo** simulation model with **probabilistic modeling** to forecast stock prices

Positions of Responsibility

Developer | **DevCom** | **IIT Bombay** [Mar '25 - Mar '26]

*Member of one of **India's largest** student developer communities, building and maintaining institute-wide digital infrastructure*

- Built and shipped **8+** features for Resobin, a productivity platform used by **10k+** students that increased engagement
- Developed the exam timetable, interactive schedule, Quick Actions, and other core features, increasing usability

Teaching Assistant | **CS101** | **Computer Programming and Utilization** [Jul '25 - Nov '25]

*Selected as **one of only two** sophomores appointed for the CS101 TA role via a programming test & technical interview*

- Led **7+** weekly lab sessions, mentoring **20+** students in C++, DSA, debugging, and algorithmic problem-solving
- Delivered support to **100+** students through help sessions covering data structures, STL, recursion and pointers

Department Academic Mentor | **Student Mentorship Program** | **IITB** [Jul '26 - Present]

*Selected as **1 of 8** DAMP Mentors through a structured SoP, peer review, and interview process from **26+** applicants*

- Mentoring **6** 2nd-year students, providing academic guidance and ensuring a smooth transition into the department
- Managing social media handles as part of the DAMP's web team, designing and publishing posts to boost outreach

Industrial Outreach Secretary | **IEORSA** | **IITB** [Jul '26 - Present]

- Driving recruiter engagement across **8+** industry domains to expand professional opportunities for **200+** IEOR students
- Strengthening IEOR's industry presence by promoting faculty expertise, research, and student excellence externally

Technical Skills

Languages & Software	Python | C++ | JavaScript | AutoCAD Fusion 360 | LaTeX| Docker | EasyEDA | Gurobi
Frameworks	LangGraph | OpenRouter | React | Django | Flask | Django REST framework
Libraries	NumPy | Pandas | Matplotlib | Scikit-learn | PyTorch | TensorFlow | HuggingFace

Key Courses Undertaken

Machine Learning	Introduction to Machine Learning | Machine Learning for Natural Language and Knowledge Graphs
Programming	Computer Programming and Utilization | Data Structures and Algorithms | Computer Networks
Mathematics	Differential Equations | Linear Algebra | Univariate and Multivariate Calculus
Operations Research	Linear Optimization | Probability and Statistics | Non-Linear and Discrete Optimization

Extra-Curricular Activities

- Selected as a Top **16** finalist in **P&G** IT Spotlight from a pool of **1,300+** applicants across leading IITs and IIITs.
- Selected among the Top **30** finalists for **HUL** Tech Foundation School from **1,000+** applicants across leading IITs.
- Secured **3rd place** among **300+** teams in SeekOut, IIT Bombay's **largest** cryptic hunt by SARC
- Led choreography for both the **winning** and **runner-up** teams in the Freshers' Dance General Championship
- Completed a trek to Naneghat Peak (**800m+** elevation), demonstrating endurance, discipline, and team spirit